



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Intelligent Knowledge Discovery Platform (IKDP)

The domain of the Project:
Generative AI

Team Mentors (and their designation):
Mr. Sreekanth S — AI Engineer

Team Members:

Ms. MANO SHRUTHI S

Period of the project

March 2026 to September 2026



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Declaration

The project titled **“Intelligent Knowledge Discovery Platform (IKDP)”** has been mentored by **Mr. Sreekanth S**, organised by SURE ProEd, from **March 2026 to September 2026**, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. MANO SHRUTHI S ----- -- Signature

Mentor's Name

Designation—Company Name

Mr. Sreekanth S

AI Engineer

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

The **Intelligent Knowledge Discovery Platform (IKDP)** is an AI-powered web application that helps researchers, students, and working professionals analyse uploaded documents, organise research materials into workspaces, and generate grounded insights, summaries, and citation-backed answers. The platform was developed using a **Retrieval-Augmented Generation (RAG)** architecture that combines a React-based frontend with a FastAPI backend, a SQLite database, a FAISS vector store, a BM25 keyword index, and the Google Gemini 2.5 large language model. Users upload PDFs, Word, spreadsheet, notebook, or source-code files; the system extracts text (with OCR support for scanned documents), chunks and embeds the content, and lets users ask questions — responses are streamed live and always include verifiable source citations. An intelligent research discovery module searches academic papers (ArXiv / EuropePMC), code repositories (GitHub), and datasets (HuggingFace), while workspaces let users attach documents, notes, resources, and datasets per project. The application is fully deployed on Render and available for online use. The project achieved all planned objectives, demonstrating a fully functional, deployed, industry-relevant generative AI solution that reduces information overload and promotes evidence-based research.



Introduction

Background and context. With the rapid growth of research publications, code repositories, and datasets, finding accurate and trustworthy information has become difficult. Generic search engines return huge lists of results, and general-purpose chatbots often give answers without verifiable sources. Students and professionals spend many hours manually reading documents, extracting insights, and organising references.

Problem statement. Researchers and students often work with large numbers of documents and research materials from different sources. Finding relevant information, keeping these materials organised, and verifying information from documents can be time-consuming and difficult. This creates a problem of scattered research information and makes it harder to efficiently access and verify knowledge when needed.

Goals of the project.

- Build a document-grounded Q&A assistant (named Orbot) that answers only from uploaded content with strict lexical verification and citations.
- Provide intelligent search across academic papers, code repositories, and datasets.
- Offer workspace-based organisation of documents, notes, and learning resources.

Scope and limitations. The platform is designed for personal and small-team research use. File uploads are limited to 25 MB per document and a wide range of supported file types; generated answers depend on the quality of the uploaded documents, and Live LLM features require a server-side Gemini API key, which is securely configured in the hosted deployment. (the hosted deployment runs with its own key).

Innovation component. The project includes several advanced RAG techniques: **hybrid search** combining dense vector search (FAISS + Sentence-Transformers) with lexical BM25 search, **contextual retrieval** that enriches chunks before embedding, **anaphora query expansion** for natural multi-turn conversations, a re-ranker to improve retrieval relevance, GraphRAG for relationship-aware retrieval, **smart citations** with page-level highlights, real-time **Server-Sent Events (SSE)** answer streaming, OCR enrichment for scanned PDFs, and automatic metadata extraction and tagging.



Project Objectives

Objectives.

- Design and build a full-stack RAG application with a modular and scalable architecture.
- Implement secure document upload, text extraction, chunking, and vector indexing.
- Develop a grounded Q&A engine that generates answers with exact source citations.
- Build a research discovery module covering papers, code, and datasets.
- Implement workspace-based project organisation with notes, documents, and resources.
- Deploy the application live and test it end-to-end.

Expected outcomes and deliverables.

- A fully functional, deployed web application (frontend and backend on Render).
- Interactive API with documentation (FastAPI /docs) and health endpoints.
- Unit-tested backend covering the core services.
- Complete source code published on GitHub with documentation.



Methodology and Results

Methods / Technology used.

- **Retrieval-Augmented Generation (RAG) pipeline:** upload → text extraction → chunking → embedding → vector store → retrieval → re-ranking → grounded LLM generation with citations.
- **Hybrid retrieval:** FAISS dense vectors plus Rank-BM25 lexical index, merged and re-ranked.
- **Contextual retrieval** and **GraphRAG** to improve answer accuracy for large document sets.
- **Streaming responses** using Server-Sent Events for a real-time chat experience.
- **Data collection approach:** documents (academic PDFs, code files, spreadsheets, notes, images) uploaded by the user; discovery data pulled from public APIs such as ArXiv, EuropePMC, GitHub, and HuggingFace.

Tools / Software used.

- Frontend: React 19, Vite, React Router v7, Framer Motion, PDF.js, React Markdown, Lucide Icons.
- Backend: FastAPI, Python 3.12, Uvicorn, Pydantic v2.
- Databases: SQLite3 (relational) and FAISS (vector store).
- AI / ML: Google Gemini 2.5, Sentence-Transformers (all-MiniLM-L6-v2), Rank-BM25, NetworkX.
- Document processing: PyMuPDF, python-docx, OpenPyXL, BeautifulSoup4, OCR service for scanned PDFs.
- Security: URL import allow-list with SSRF protection, file-type and size validation, rate limiting.
- DevOps: Docker, Git & GitHub, Render (Web Service + Static Site).

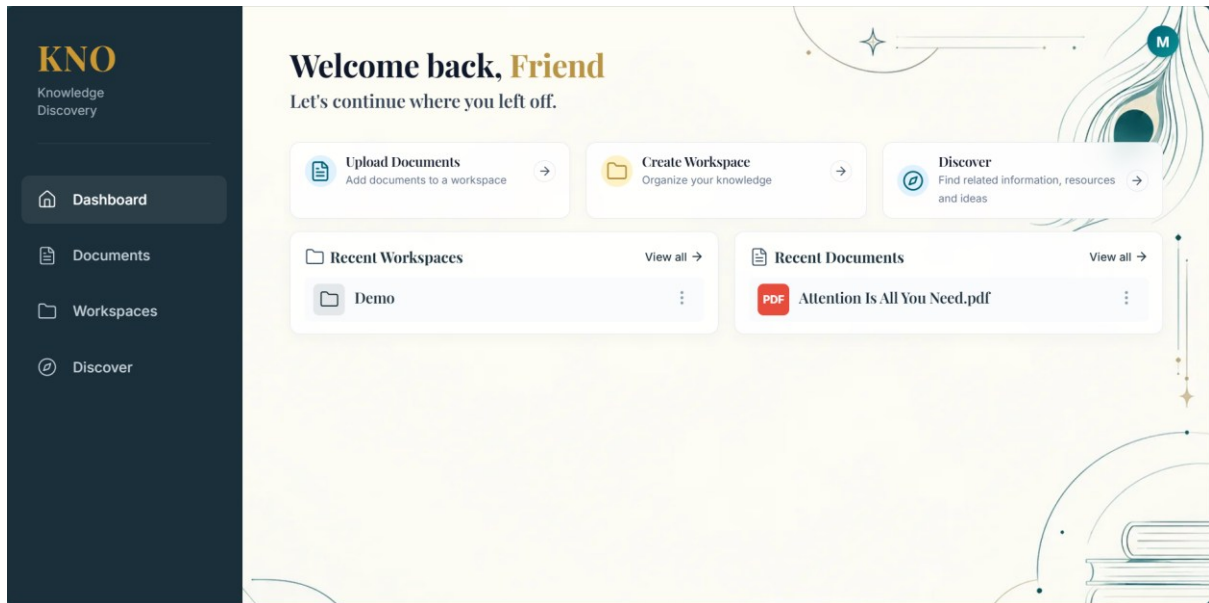
Project Architecture. The system follows a clean client–server architecture:

- **Frontend (React + Vite)** provides the dashboard, document upload, chat interface, discover page, and workspace pages. It talks to the backend through a REST API and streams answers over SSE.
- **Backend (FastAPI)** exposes REST endpoints for documents, workspaces, notes, resources, search, question answering, summaries, insights, and discovery. It enforces CORS, rate limits, and input validation.
- **AI layer:** the Orbot RAG engine (query understanding, hybrid search, re-ranking, grounding, citation generation) and the LLM service that supports provider fallback.
- **Storage:** SQLite holds documents, workspaces, conversations, chunks, and analysis cache; FAISS and a BM25 index store the searchable content; Uploaded files are stored in the application's upload storage.
- **Project GitHub Link:** <https://github.com/sure-trust/MANO-SHRUTHI-S-g6-gen-ai>



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Results. The application is deployed and working: **Live app:** <https://ikdp-frontend.onrender.com/>



Backend health: <https://ikdp-backend.onrender.com/api/health>.



End-to-end testing confirmed document ingestion, hybrid retrieval, streamed citation-backed answers, OCR enrichment, and research discovery all function as designed.

Social / Industry Relevance of the Project

The project directly addresses the problem of information overload and unverifiable AI answers. Researchers, students, and professionals gain a trustworthy assistant that cites every claim, saving hours of manual reading. The platform is relevant to academia (literature review, research gap analysis), the AI industry (RAG-based knowledge assistants), and corporate settings (internal document Q&A). Because it is built on widely used open-source technologies and a modular RAG pipeline, it can be extended and reused across industries.



Ms. MANO SHRUTHI S — Learnings.

- **Technology:** building full-stack Generative AI applications; Python and FastAPI for backend APIs; React and Vite for the frontend; SQLite for relational data; FAISS and BM25 for vector and keyword search; the complete RAG pipeline (chunking, embeddings, retrieval, re-ranking, generation); Google Gemini 2.5 prompting; OCR; REST and SSE; deployment on Render with Docker; and Git/GitHub version control.
- **Management:** requirement analysis, modular project planning, structured coding, testing with automated test suites, documentation, and iterative bug-fixing.
- **Soft skills:** problem-solving, self-learning, attention to detail, time management, and technical documentation.

Ms. MANO SHRUTHI S — Overall experience. Working on IKDP was a valuable end-to-end internship experience. I learned how a real-world AI product is designed, built, tested, and deployed, and I gained confidence in implementing advanced RAG features such as hybrid retrieval, contextual retrieval, grounding with citations, and streaming chat. The project strengthened both my technical depth and my ability to deliver a complete, working product.



Conclusion and Future Scope

Recap. All objectives were met: IKDP successfully uploads and organises documents, answers questions with verifiable citations, discovers research across multiple sources, and is deployed live. The platform demonstrated a complete RAG workflow built with modern, industry-standard tools.

Future scope.

- Multi-user authentication and collaborative workspaces.
- Support for more languages and additional file formats, including audio/video transcription.
- Fine-tuning the embedding and re-ranking models on domain-specific corpora.
- Integration with more data sources (e.g., patents, news, internal intranets).
- Content safety filters, model-agnostic provider addition, and usage analytics dashboards.
- Mobile/desktop app versions and offline document analysis.